Supplementary Materials for

War Mobilization and Economic Development: World War II and Structural Transformation in India

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This document provides additional data documentation and maps that supplement the Online Appendix for "War Mobilization and Economic Development: World War II and Structural Transformation in India."

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1 Census Data: Year-by-Year Documentation

This section provides detailed documentation for each Census year on the specific tables from which population and occupation data were extracted, the definition of "workers" used in that year, adjustments for unproductive workers, and other year-specific data processing details. General definitions of agricultural, industrial, and service sector workers are provided in the Online Appendix.

1.1 1911

The occupational data for the 1911 census was extracted from Table XV (Occupation or Means of Livelihood) from each of the Indian provinces, states and agencies. For the census year 1911, we define workers to be "actual workers" ("Total Workers and Dependents" minus "Dependents"). We define workers in sub-class IX (Persons living on their own income) and sub-class XII (Unproductive) from this table to be unproductive workers. The data for population is extracted from Table I (Area, Houses and Population) from each of the province censuses.

1.2 1921

The occupational data for the 1921 census was extracted from Table XVII (Occupation or Means of Livelihood, Part B - General Table – Details for Districts) from each of the provincial census PDFs. The data on population is from Table I (Area, Houses and Population). We define persons principally living on their own income (order 51), inmates of jails, asylums and almshouses (order 54), beggars, vagrants, prostitutes (order 55) and other unclassified non-productive industries (order 56) from table XVII as unproductive workers and remove them from the employment figures. We define workers to be "Actual workers" in a manner similar to 1911.

1.3 1931

The occupational data for the 1931 census was extracted from Table X (Occupation or Means of Livelihood) from each of the Indian provinces, states and agencies. For the census year 1931, we define "workers" as the sum of principal earners and earning dependents. The 1931 census report describes the sum of principal earners and earning dependents in effect to be equivalent to the workers as recorded by 1921 census. We define workers in occupation groups 185 (Persons living principally on their income), 192 (Inmates of jails, asylums and alms houses), 193 (Beggars and vagrants) and 195 (Other unclassified non-productive industries) as unproductive workers and remove them from employment figures. We consider those working as procurers and prostitutes (group 194) as employed. (In the previous two censuses, this occupation was not reported separately.)

1.4 1951

For the census year 1951, we extract data on livelihood classes and population from Economic Table B-I (Livelihood Classes and Sub-classes) and data on unproductive workers from the table in the preamble preceding Economic Table III for each of the Indian states. The introduction before Table BIII provides district-level information on various types of unproductive workers. In these tables, the livelihood classes were divided into three sub-classes, namely self-supporting persons, non-earning dependents and earning dependents. For the year 1951, we define workers as the sum of self-supporting persons and earning dependents. For this year, data in table B-I is reported for eight livelihood classes (e.g., Agricultural Class I: cultivators of land wholly owned; Agricultural Class II: cultivators of land wholly or mainly unowned; etc.) "Primary industries" (division 0) such as fishing, hunting etc are classified by the census as included as a part of the industry sector, so we re-assign these workers to the agriculture sector. We extract district level data on division 0 from Table B-III (Employers, Employees and Independent Workers in Industries and Services by Division and Sub-divisions).[1] For the year 1951, data on Jammu and Kashmir districts are not reported and are hence not a part of our analysis for the year 1951.

1.5 1961

For the year 1961, we extract data on workers and population from Table B-1 (Workers and Non-workers Classified by Sex and Age Group.) for each of the Indian states apart from Madras and North-Eastern Frontier Agency. For these two states, we were not able to locate PDFs containing the B-I economic table. Therefore, for these two states, we extracted information from Union Primary Census Abstracts. In the 1961 census, workers and non-workers as aggregate categories are reported but no distinction is made between principal earners and working dependents. We use the definition of "workers" in the 1961 census. From 1961 census onwards, non-workers are reported separately. No further adjustments are required to remove unproductive workers.

Similar to 1951, primary industries like fishing, hunting and so on are reported along with mining, quarrying and hunting. We re-assign the workers in fishing, hunting, and the like (referred to in the census as Division 0) from industry to agriculture. We extract the data on division 0 workers from table B-IV Part C in state censuses. (Industrial Classification by Sex and Class of Worker of Persons at Work in Non-household Industry, Trade, Business, Profession or Services). For Madras, we extract the data for workers in division 0 from Table B-V (Occupational Classification by Sex of Persons at Work Other than Cultivation).

[1] Table B-III provides information on self-supporting persons only, and no data on earning dependents.

1.6 1971

For the year 1971, we extract data on population and workers from Table B-I Part A (Workers and Non-workers According to Main Activity Classified by Sex and Age Groups) for each of the Indian states (except Mysore). For Mysore, we were not able to locate the General Economic Tables PDF, and hence extract data from the primary census abstract. The data on primary industries is reported on its own from the 1971 census onwards and no further adjustments are necessary.

1.7 1981

From 1981 census and onwards data on two types of workers are reported, namely main workers and marginal workers. Main workers are broadly defined as those who work for over 6 months in a year and marginal workers are defined as those who work for fewer than 6 months in a year. We define workers in a certain sector as the sum of both marginal and main workers engaged in that sector. For the 1981 census year, the data on population and main workers is extracted from table B3 (Main Workers Classified by Industrial Category, Age and Sex) and data on marginal workers is extracted from table B7 (Marginal Workers by Industrial Category, Age and Sex) for each of the Indian states (apart from Assam and Arunachal Pradesh.[2])

For the census years 1911-1981, whenever number of workers in industry i is not reported, we assume it to be 0.

1.8 1991

From 1991 census year onwards, the data tables are available in electronic format on the Indian census website, and so did not require data entry. For 1991, we download state-wise tables B-2(S) (Main Workers Classified by Industrial Category, Age and Sex), B-2(F) (Main Workers Other Than Cultivators And Agricultural Labourers Classified By Industrial Category, Age And Sex) and B-6(F) (Marginal Workers Classified By Industrial Category Of Work And Main Activity) from the census website. B-2(S) tables provide information on population, total main workers, main workers in agriculture, cultivation, manufacturing in household industry and sum of main workers in other industrial categories. B-2(F) tables provide us more detailed information on main workers in other industrial categories like mining and quarrying, construction and so on. For the census year 1991, data on Jammu and Kashmir districts is not reported. We do not include Jammu and Kashmir in our analysis for the year 1991.

[2]We were not able to locate the General Economic Tables for these states. In any case, these states are not included in our analyses due to their proximity to the Japanese advance in Burma and the subsequent orientation of war production away from the northeast and Bengal.

1.9 2001

For 2001, we download data tables B-04 (Main Workers Classified by Age, Industrial Category and Sex (All)) for data on main workers, B-06 (Marginal Workers Classified by Age, Industrial Category and Sex) for data on marginal workers and district-wise population data from the Indian census website for population data.

1.10 2011

For the census year 2011, we download tables B-04 (Main Workers Classified by Age, Industrial Category and Sex (Total)) for data on main workers and B-06 (Marginal Workers Classified by Age, Industrial Category and Sex (Total)) for data on marginal workers for each of the Indian states from the Indian census website. We also download table B-18 (Industrial Classification of Main Workers and Marginal Workers other than Cultivators and Agricultural Labourers by Sex and by Section, Division and Class) for detailed data on workers in employed in different occupations. We use this data to determine workers in various service sub-categories as well as get national level estimates of workers engaged in different occupations.

2 District Concordances: Detailed Documentation

We use the crosswalk between modern Indian sub districts and historic districts constructed in Fenske et al. (2025) to form 'districts' that stay consistent between any two census decades.[3] We make minor changes to this crosswalk for the purposes of matching it with census as detailed below. Henceforth, we refer to this crosswalk as the "geographic boundaries concordance". For each census year, we map the districts as reported in census data with the districts reported in the geographic boundaries concordance. This allow us to assign economic data to geographically consistent districts over time.

Districts that were under the rule of non-British colonial powers (Portugal and France) are not present in any pre-independence censuses and hence do not match with the geographic concordance. Examples of such districts are Yanam, Pondicherry, Mahe and Karaikal, which were a part of French India before 1950. Other such districts are Goa, Dadra and Nagar Haveli and Daman and Diu, which were under Portuguese rule until 1961.

Districts that were a part of India before independence but are in modern day Pakistan or Bangladesh also drop out since the geographic boundaries concordance is only for modern day Indian districts. Similarly, we did not find census data for small princely states in Central India Agency for 1911 and 1921. As a result these are not included in our pre-trend sample. If the census data is missing for a district

[3] This crosswalk (gadmtohist_indiaonly.xlsx) is publicly available and can be downloaded from https://warwick.ac.uk/fac/soc/economics/staff/jefenske/data/.

or a part of district, then it drops out of our sample and is depicted in grey in maps in the Online Appendix.

2.1 Modifications to geographic boundaries concordance

We rename districts that appear in different states to have an abbreviation of the state as a suffix. For example, if two different districts are called Bilaspur, we rename these districts to Bilaspur (cp berar) and Bilaspur (Punjab) to reflect that these are two separate districts in two distinct provinces.

Mapping princely states to political agencies. Under British rule, there were various political agencies. These are autonomous or semi-autonomous units and are mostly comprised of princely states. The geographic concordance provides information on these princely states and how they map to pre- and post-independence Indian districts. However, the Census reports information on political agencies (an aggregate unit that comprises of these princely states). To know how the princely states map to political agencies in pre-independence India, we visually compare 1911, 1921 and 1931 maps from the Indian Census atlas with census province maps (mainly for Bombay presidency) to concord these princely states with political agencies. For the 1911 Census, while the political agencies are directly reported, for 1921 and 1931 Censuses, information for some of the main princely states is reported and data for the other princely states is reported as "Rest of political agency". So for 1911 we concord the princely states with political agencies. For 1921 and 1931, we concord princely states (which are not directly reported in the census) with "Rest of the appropriate political agency". In 1911, 1921 and 1931 some princely states are in the Indian atlas but they are missing in the geographic boundaries concordance. In these cases, by visual comparison we assign these princely states to political agencies in pre-independence years and map them to 2011 districts by relying on district census handbooks and official district webpages. We provide more detailed explanations below.

1911:

- Balasinor, Lunwara, Sunth, Bariya, Chota Udepur, Pandu Mewas and Rajpipla in the geographic boundaries concordance map to Rewa Kantha Agency in the 1911 census. Kadana, Sanjeli and Sankheda Mewas are in Census atlas but not in geographic boundaries concordance. We consider these to be a part of Rewa Kantha Agency in 1911. We map these three princely states to modern day Indian districts by relying on census webpages. According the Indian webpage for Dahod district, Sanjeli is one of the sub-districts of Dahod. Similarly, according to Indian government webpage for Mahisagar district (carved out of Kheda and Panch Mahals), Kadana is one of the sub-districts of Mahisagar and according to the 2001 census handbook, Sankheda is one of the sub-districts of Vadodara.
- Santalpur, Wardhi, Radhanpur, Wao, Tharad, Diodar, Kankrej and Palanpur map to Palanpur Agency

- Porbandar, Junagarh, Bhavnagarh, Palitana, Lathi, Jasdan, Kotra Sangani, Bagasra, Jetpur, Gonda, Nawanagar, Dhrol, Khairasra, Rajkot, Morvi, Wankaner, Malia, Dhrangadra, Eastern Kathiawar Agency and Jafrabad map to Kathiawar agency. Bantwa, Bagasra and Virpur are in Census atlas but not in geographic concordance. We map these to Kathiawar agency in 1911. According to the 2001 district census handbook for Junagadh, Bantwa princely state is in Junagadh. Similarly, we see that there is a village called Virpur in Junagadh princely state. We geographically confirm the location by visually comparing Virpur princely state in 1911 with modern day Junagadh district. The 2011 district handbook of Amreli district mentions that the district comprises of former princely state Bagasara.
- Jat maps to Bijapur Agency.
- Mewas maps to Khandesh Agency.
- Oundh and Phaltan map to Satara Agency.
- Bansdra, dang and Dharampur map to Surat Agency.
- Jamkhandi, Mudhol, Ramdurg, Sangli, Miraj map to Southern Marath Jagirs.
- Athmallik, Pal Lahara, Kalahandi, Sonpur, Baud, Daspalla, Baramba, Dhenkanal, Hindol, Talcher, Rairakhol, Bamra, Keonjhar, Mayurbhanj, Nilgiri and Patna (different from Bihar capital) map to Orissa feudatory states. Ranpur, Nayangarh, Khandpara, Narsinghpur, Tigiria, Athgarh are present in the census atlas but missing in the geographic boundaries concordance. In 1911, we map these to be orissa tributary states. In 2011, we rely on district gazattes and handbooks to map these princely states. According to the Orissa District Gazetteer from 1993, Narsinghpur, Athgarh and Tigria are mentioned as sub-divisions of the Cuttack district and according to the 2011 census handbook, Ranpur, Nayagarh and Khandapada (Khandpara) are blocks (administrative sub-divisions) in Nayagarh district.
- Gangpur, Bonai, Saraikela, Kharsawan map to Chota Nagpur States.

1921:

- Athmallik, Pal Lahara, Kalahandi, Sonpur, Baud, Daspalla, Baramba, Dhenkanal, Hindol, Talcher, Rairakhol, Bamra, Keonjhar, Mayurbhanj, Gangpur, Nilgiri and Patna (different from Bihar capital) map to Orissa states. Similar to 1911, some princely states (Ranpur, Nayangarh, Khandpara, Narsinghpur, Tigiria, Athgarh) are in the Indian census atlas but missing from the geographic boundaries concordance. We treat them in same manner as 1911.
- Santalpur, Wardhi, Radhanpur, Wao, Tharad, Diodar, Kankrej map to Rest of Palanpur Agency.
- Balasinor, Lunwara, Sunth, Baria, Chota Udepur, Pandu Mewas map to Rest of Rewa Kantha Agency.

- Bansda, Dangs, Dharampur map to Surat Agency.
- Porbandar, Palitana, Lathi, Jasdan, Kotda Sanghani, Bagasra, Jetpur, Dhrol, Khairasra, Rajkot, Morvi, Wankaner, Malia, Dhrangadra, Eastern Kathiawar Agency and Jafrabad map to Rest of Kathiawar. Bantwa, Bagasra and Virpur are missing in the Indian census atlas but missing from the geographic concordance. We treat them in the same manner as 1911.
- Jamkhandi, Mudhol, Ramdurg, Miraj map to other Southern Maratha Country states.
- We map the remaining unassigned princely states to "Other states in the presidency proper" row in census. The princely states that are mapped in this manner are Oundh, Phaltan, Janjira, Jawhar, Surgana, Jath and Akalkot.

1931:

- Deodar, Kankarej, Santalpur map to Rest of Banaskantha.
- Pandu Mewas maps to Rest of Reva Kantha.
- Saraikela maps to Chota Nagpur States.
- Athmallik, Pal Lahara, Kalahandi, Sonpur, Baud, Daspalla, Baramba, Dhenkanal, Hindol, Talcher, Rairakhol, Bamra, Keonjhar, Gangpur, Mayurbhanj, Nilgiri and Patna (different from Bihar capital) map to Orissa feudatory states. Ranpur, Nayangarh, Khandpara, Narsinghpur, Tigiria, Athgarh are in the Indian census atlas but missing from the geographic concordance. We treat them in the same manner as 1911.

1991: In 1991, states are coded incorrectly for Raigarh in the geographic boundaries concordance. There are two Raigarh districts in India, one in Maharashtra and the other in Madhya Pradesh. For 1991 only, Raigarh in Madhya Pradesh is erroneously coded to be from Maharashtra. We make this correction in the geographic boundaries concordance to correctly map 1931 districts to 1991 districts.

3 Modifications and Missing Data in Censuses

We make minor modifications to censuses for different years which we document in this section. We also make spellings of different districts consistent between the census data and the geographic boundaries concordance.[4]

1931: Miraj appears as Miraj (junior) and Miraj (senior) in census data. Dewas appears as Dewas (junior) and Dewas (senior) in census data. We sum up the occupation and population data for Miraj (junior) and Miraj (senior) data to obtain the data for Miraj as a whole and similarly for Dewas. We have to do this because we do not see separate entries for senior and junior in the geographic boundaries concordance.

[4]For details, see our census-Fenskeconcordance crosswalk files.

After visually comparing map of Bombay Presidency from pre-independence census reports and the census atlas, we determine Wardhi and Varahi to refer to the same region. We map "d. s. vala mulu surang of jetpur (pithadia)" and "d. s. vala rawat ram of jetpur (bilkho)" in the 1931 census with Jetpur in the 1931 geographic boundaries concordance.

In addition to territories under French or Portuguese rule, we do not map a few districts that are in the pre-independence geographic concordance to the relevant census year either because we could not locate relevant detailed census data for these districts or detailed occupational data was not collected for these districts. Examples of such districts for 1931 are given below.

- Rampura (Indore)
- Kukshi (Indore)
- Nemawar (Indore)
- Rann of Kutch
- Naga Tribes
- Singhpos
- Districts from Rajputana Agency
- Lakshadweep Islands

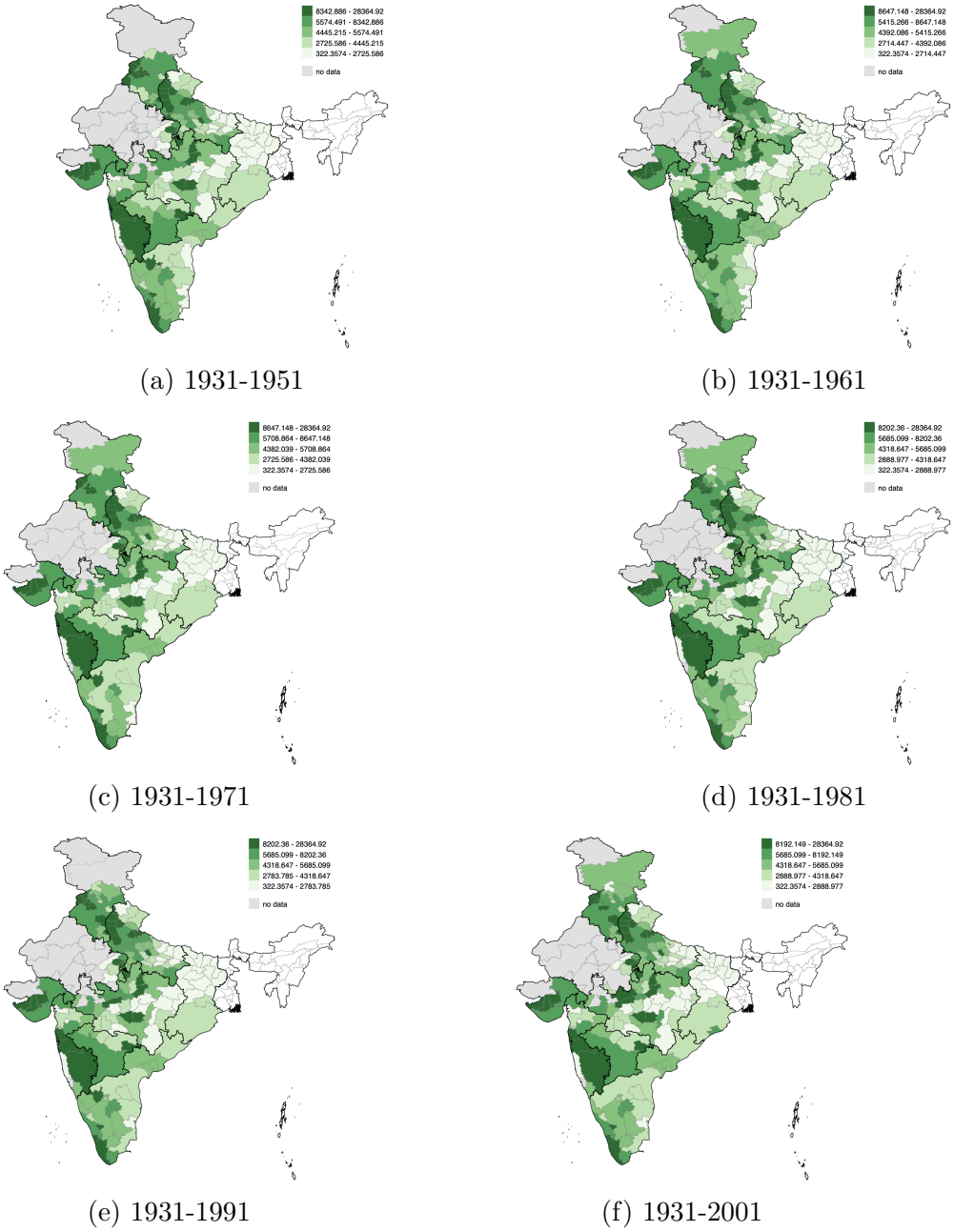
In addition, for 1911 and 1921, we were not able to locate census data on districts in Central India Agency. As a result, these districts drop out of our pre-trend analysis.

1981. For the year 1981, we were not able to locate census data for Assam and Arunachal Pradesh states. However, districts in these states are not a part of our main sample.

2010 and 2011 Concordance. For the years 2010 and 2011, we map all of Delhi National Capital Territory region to 1931 Delhi. In the geographic boundaries concordance, 1931 Delhi is mapped to just West Delhi in 2010 and 2011.

4 Additional Maps: Post-War Sample Spatial Variation

Figure 1: Spatial Variation in Shift-Share in Post-War Samples



(a) 1931-1951 (b) 1931-1961

(c) 1931-1971 (d) 1931-1981

(e) 1931-1991 (f) 1931-2001

Notes: In each of the maps, districts shown are consistent geographic units between 1931 and different post-war decadal census. Light grey lines demarcate district borders. Black lines demarcate larger-scale "regions" (author defined) of contiguous groups of districts (for estimation of region * time fixed effects). Green shading represents value of shift-share variable in real 2011 Indian rupees (INR). Grey shading indicates districts for which we cannot calculate the shift-share variable due to availability of Indian Census data. Districts in white (in northeast) are not included in analysis, due to proximity to Japanese military advance in Burma. Andaman and Nicobar Islands also not included in analysis since they were occupied by Japan during the war.

References

Fenske, J., B. Gupta, and C. Neumann (2025): "Missing women in colonial india," *Economic History Review*, 78.